Lecture Notes

Lecture 9

Adjoints; Self-Adjoint, Unitary, and Normal Operators

The Riesz theorem of Lecture 8 turned vectors into functionals. Applied to operators it produces the *adjoint* $T^\dagger$, the abstract version of the conjugate transpose, defined by $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$. With the adjoint in hand we can single out the three families of operators that organize the finite-dimensional operator model used in quantum mechanics: the *self-adjoint* operators $T^\dagger = T$, whose real eigenvalues label the possible outcomes of an ideal projective measurement; the *unitary* operators $U^\dagger = U^{-1}$, which preserve inner products and implement closed-system time evolution and unitary symmetries (other symmetries may be antiunitary); and the *normal* operators $T^\dagger T = TT^\dagger$, the widest class that admits an orthonormal eigenbasis in finite-dimensional complex spaces. This lecture builds the adjoint and proves the spectral facts—real eigenvalues, orthogonal eigenvectors, unit-circle spectra—that the spectral theorem of Lecture 10 will complete.

Learning objectives

By the end of this lecture you will be able to:

- Compute the adjoint $T^\dagger$ and use $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$ and its algebra.
- Classify operators as self-adjoint, unitary, or normal.
- Prove the core facts: real eigenvalues, unit-circle spectrum, orthogonal eigenvectors.
- In the finite/discrete pure-point model, identify observables, unitary versus antiunitary symmetries, and the class the finite spectral theorem diagonalizes.

1 The adjoint

Definition 9.1 (Adjoint). Let $T \in \mathcal{L}(V)$ on a finite-dimensional inner product space. The *adjoint* $T^\dagger$ is the unique operator satisfying

$$\langle T^\dagger u, v\rangle = \langle u, Tv\rangle \qquad \text{for all } u, v \in V.$$

Existence and uniqueness are Riesz in disguise: for fixed u, the map $v \mapsto \langle u, Tv\rangle$ is a linear functional, so equals $\langle w, v\rangle$ for a unique w_u. Uniqueness also proves that these representatives depend linearly on u: $w_{u_1+u_2} = w_{u_1} + w_{u_2}$ because both sides represent $v \mapsto \langle u_1 + u_2, Tv\rangle$. For $a \in \mathbb{F}$, the functional for au is $\phi_{au} = \overline{a}\,\phi_u$, while $\langle aw_u, v\rangle = \overline{a}\,\langle w_u, v\rangle$; uniqueness therefore gives $w_{au} = aw_u$. Thus setting $T^\dagger u = w_u$ really does define a linear operator. Equivalently, taking conjugates, $\langle Tu, v\rangle = \langle u, T^\dagger v\rangle$: the adjoint moves an operator across the inner product from one slot to the other.

Proposition 9.2 (The adjoint is the conjugate transpose). In an orthonormal basis, $\mathcal{M}(T^\dagger) = \mathcal{M}(T)^\dagger$: the matrix of the adjoint is the conjugate transpose of the matrix of T.

Proof. With orthonormal $e_1, \ldots, e_n$, the (j,k) entry of T is $T_{jk} = \langle e_j, Te_k\rangle$. Hence

$$(T^\dagger)_{jk} = \langle e_j, T^\dagger e_k\rangle = \langle Te_j, e_k\rangle = \overline{\langle e_k, Te_j\rangle} = \overline{T_{kj}},$$

which is exactly the conjugate transpose. ∎

Proposition 9.3 (Algebra of adjoints). For $S, T \in \mathcal{L}(V)$ and $a \in \mathbb{F}$: $(S+T)^\dagger = S^\dagger + T^\dagger$, $(aT)^\dagger = \overline{a}\,T^\dagger$, $(ST)^\dagger = T^\dagger S^\dagger$, $(T^\dagger)^\dagger = T$, and $I^\dagger = I$. If T is invertible then $(T^{-1})^\dagger = (T^\dagger)^{-1}$.

Proof. Each follows by checking the defining relation. For the product, $\langle (ST)^\dagger u, v\rangle = \langle u, STv\rangle = \langle S^\dagger u, Tv\rangle = \langle T^\dagger S^\dagger u, v\rangle$ for all v, so $(ST)^\dagger = T^\dagger S^\dagger$. The conjugate on a comes from conjugate-linearity in the first slot. ∎

Theorem 9.4 (Null space and range). For $T \in \mathcal{L}(V)$, $\operatorname{null} T^\dagger = (\operatorname{range} T)^\perp$ and $\operatorname{range} T^\dagger = (\operatorname{null} T)^\perp$.

Proof. $u \in \operatorname{null} T^\dagger$ iff $T^\dagger u = 0$ iff $\langle T^\dagger u, v\rangle = 0$ for all v iff $\langle u, Tv\rangle = 0$ for all v iff $u \perp \operatorname{range} T$. That is the first identity. Apply it now to $T^\dagger$: $\operatorname{null} T = (\operatorname{range} T^\dagger)^\perp$, since $(T^\dagger)^\dagger = T$. Taking orthogonal complements and using the finite-dimensional identity $W^{\perp\perp} = W$ gives $(\operatorname{null} T)^\perp = \operatorname{range} T^\dagger$, the second identity. ∎

Example 9.5 (An adjoint by conjugate transpose). $A = \left(\begin{smallmatrix} 1 & 2-i \\ 3i & 4 \end{smallmatrix}\right)$ has adjoint $A^\dagger = \left(\begin{smallmatrix} 1 & -3i \\ 2+i & 4 \end{smallmatrix}\right)$: transpose, then conjugate. Since $A^\dagger \neq A$ and $A^\dagger \neq A^{-1}$, this operator is neither self-adjoint nor unitary. ◁

Example 9.6 (Adjoint of the shift). The right shift $T(z_1, \ldots, z_n) = (0, z_1, \ldots, z_{n-1})$ on $\mathbb{C}^n$ has adjoint the left shift $T^\dagger(z_1, \ldots, z_n) = (z_2, \ldots, z_n, 0)$: with the boundary convention $u_0 = 0$, from $\langle Tu, v\rangle = \sum_k \overline{u_{k-1}} v_k = \sum_j \overline{u_j} v_{j+1} = \langle u, T^\dagger v\rangle$ the adjoint reverses the shift direction. ◁

Example 9.7 (Adjoint of a product reverses order). For $S = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ and $T = \left(\begin{smallmatrix} 1 & 0 \\ i & 0 \end{smallmatrix}\right)$, $ST = \left(\begin{smallmatrix} i & 0 \\ 0 & 0 \end{smallmatrix}\right)$, so $(ST)^\dagger = \left(\begin{smallmatrix} -i & 0 \\ 0 & 0 \end{smallmatrix}\right) = T^\dagger S^\dagger$. Adjunction reverses the order of a product, just as inversion does. ◁

Example 9.8 (The defining relation, checked). With $T = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $u = (1,0)$, $v = (0,1)$: $\langle u, Tv\rangle = \langle (1,0), (1,0)\rangle = 1$, while $T^\dagger u = (0,1)$ gives $\langle T^\dagger u, v\rangle = \langle (0,1), (0,1)\rangle = 1$. The two sides of $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$ agree, as the definition requires. ◁

Remark 9.9 (Detecting the zero operator). $T = 0 \iff T^\dagger = 0 \iff T^\dagger T = 0$. The last equivalence is handy because $T^\dagger T$ is self-adjoint with $\langle v, T^\dagger T v\rangle = \|Tv\|^2$, so $T^\dagger T = 0$ forces $Tv = 0$ for every v.

Corollary 9.10 (Injectivity and surjectivity). T is injective iff $T^\dagger$ is surjective, and T is surjective iff $T^\dagger$ is injective; hence T is invertible iff $T^\dagger$ is.

Proof. T injective $\iff \operatorname{null} T = \{0\} \iff (\operatorname{range} T^\dagger)^\perp = \{0\} \iff \operatorname{range} T^\dagger = V$ by Theorem 9.4, i.e. $T^\dagger$ surjective. Exchanging T and $T^\dagger$ gives the other statement. ∎

2 Self-adjoint operators

Definition 9.11 (Self-adjoint). $T \in \mathcal{L}(V)$ is *self-adjoint* (*Hermitian*) if $T^\dagger = T$, i.e. $\langle Tu, v\rangle = \langle u, Tv\rangle$ for all u, v. Its matrix in an orthonormal basis is a Hermitian matrix, $A^\dagger = A$.

In the finite/discrete pure-point model these operators encode observables. Two facts make them useful there.

> **Theorem 9.12 (Real eigenvalues).** Every eigenvalue of a self-adjoint operator is real.

Proof. Let $Tv = \lambda v$ with $v \neq 0$. Then $\lambda\|v\|^2 = \langle v, Tv\rangle = \langle Tv, v\rangle = \overline{\lambda}\,\|v\|^2$, using self-adjointness in the middle step. Since $\|v\|^2 \neq 0$, $\lambda = \overline{\lambda}$ is real. ∎

Both of the next two theorems turn on one small fact, which is worth isolating rather than smuggling into a proof.

> **Lemma 9.13 (A vanishing quadratic form).** Let $S \in \mathcal{L}(V)$ satisfy $\langle v, Sv\rangle = 0$ for every $v \in V$. If $\mathbb{F} = \mathbb{C}$, then $S = 0$. If $\mathbb{F} = \mathbb{R}$, the same follows provided S is self-adjoint.

Proof. Expanding $0 = \langle u + v, S(u + v)\rangle$ and using $\langle u, S\rangle u = \langle v, S\rangle v = 0$ gives

$$\langle u, Sv\rangle = -\langle v, Su\rangle \qquad \text{for all } u, v. \tag{9.1}$$

Over $\mathbb{C}$, replace u by iu in (9.1). The left side becomes $\overline{i}\,\langle u, S\rangle v = -i\langle u, S\rangle v$ (conjugate-linear first slot) and the right side $-i\langle v, S\rangle u$ (linear second slot), so $\langle u, S\rangle v = \langle v, S\rangle u$. Together with (9.1) this forces $\langle u, S\rangle v = 0$ for all u, v, hence $Sv = 0$ for every v. Over $\mathbb{R}$ with S self-adjoint, $\langle v, S\rangle u = \langle Sv, u\rangle = \langle u, Sv\rangle$, and (9.1) reads $\langle u, S\rangle v = -\langle u, S\rangle v$, giving the same conclusion. ∎

> **Theorem 9.14 (Self-adjoint via a real quadratic form).** For $\mathbb{F} = \mathbb{C}$, an operator T is self-adjoint if and only if $\langle v, Tv\rangle \in \mathbb{R}$ for every v.

Proof. If $T^\dagger = T$ then $\langle v, Tv\rangle = \langle Tv, v\rangle = \overline{\langle v, Tv\rangle}$ is real. Conversely, if $\langle v, Tv\rangle$ is always real then $\langle v, Tv\rangle = \overline{\langle v, Tv\rangle} = \langle Tv, v\rangle = \langle v, T^\dagger v\rangle$, so $\langle v, (T - T^\dagger)v\rangle = 0$ for all v, so $T = T^\dagger$ by Lemma 9.13. ∎

> **Theorem 9.15 (Orthogonal eigenvectors).** Eigenvectors of a self-adjoint operator for distinct eigenvalues are orthogonal.

Proof. Let $Tu = \alpha u$, $Tv = \beta v$ with $\alpha \neq \beta$ (both real). Then

$$\alpha\langle u, v\rangle = \langle Tu, v\rangle = \langle u, Tv\rangle = \beta\langle u, v\rangle,$$

using $\langle Tu, v\rangle = \overline{\alpha}\langle u, v\rangle = \alpha\langle u, v\rangle$ on the left. Thus $(\alpha - \beta)\langle u, v\rangle = 0$, forcing $\langle u, v\rangle = 0$. ∎

In the finite ideal-projective model, a measurement of a self-adjoint observable returns one of its real eigenvalues. Distinct eigenvalues have orthogonal eigenspaces and can therefore be distinguished by the corresponding spectral measurement. Continuous-spectrum observables require the projection-valued measures of Lecture 14 instead of ordinary normalizable eigenvectors.

Example 9.16 (The Pauli matrices are observables). Each Pauli matrix $\sigma_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\sigma_y = \left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$, $\sigma_z = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$ equals its own conjugate transpose, so all three are self-adjoint with real eigenvalues ± 1—the two outcomes of a spin measurement—and orthogonal eigenstates. ◁

Example 9.17 (Orthogonal eigenvectors in action). The Hermitian matrix $A = \left(\begin{smallmatrix} 2 & i \\ -i & 2 \end{smallmatrix}\right)$ has real eigenvalues $1, 3$ with eigenvectors $(1, i)$ and $(1, -i)$. Their inner product $\langle(1, i), (1, -i)\rangle = \overline{1}\cdot 1 + \overline{i}\cdot(-i) = 1 + (-i)(-i) = 1 - 1 = 0$ confirms Theorem 9.15. ◁

Example 9.18 (An orthonormal eigenbasis). Normalizing those eigenvectors gives the orthonormal pair $e_1 = \frac{1}{\sqrt{2}}(1, i)$, $e_2 = \frac{1}{\sqrt{2}}(1, -i)$, in which the operator becomes $\operatorname{diag}(1, 3)$. A self-adjoint operator is thus diagonal in an orthonormal basis—the spectral theorem of Lecture 10, previewed. ◁

Example 9.19 (A 3×3 self-adjoint operator). The real symmetric matrix $\left(\begin{smallmatrix} 14 & -13 & 8 \\ -13 & 14 & 8 \\ 8 & 8 & -7 \end{smallmatrix}\right)$ equals its transpose, hence is self-adjoint, with real eigenvalues $27, 9, -15$ and orthonormal eigenvectors $\frac{1}{\sqrt{2}}(1, -1, 0)$, $\frac{1}{\sqrt{3}}(1, 1, 1)$, $\frac{1}{\sqrt{6}}(1, 1, -2)$. In that orthonormal basis it is $\operatorname{diag}(27, 9, -15)$—real spectrum, orthogonal eigenstates, exactly as the theory predicts. ◁

Remark 9.20 (The real case). A real symmetric matrix is self-adjoint for the real dot product, so it too has real eigenvalues and an *orthogonal* eigenbasis—the real spectral theorem. This is why the principal axes of an ellipsoid, the moments of inertia of a rigid body, and the normal modes of a coupled system come out mutually perpendicular.

Example 9.21 (Diagonalizing σ_x). $\sigma_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has real eigenvalues ± 1 with orthonormal eigenvectors $|\pm\rangle = \frac{1}{\sqrt{2}}(1, \pm 1)$. In the unitary basis $U = \big(|+\rangle\ |-\rangle\big)$,

$$U^\dagger \sigma_x U = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \sigma_z,$$

so the Hadamard change of basis turns σ_x into σ_z: a self-adjoint operator becomes a real diagonal matrix in its orthonormal eigenbasis. ◁

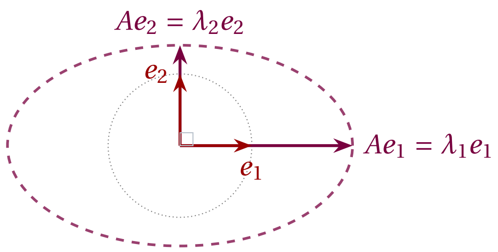

Figure 1: A self-adjoint operator stretches along an *orthogonal* eigenbasis $e_1 \perp e_2$ by real factors λ_1, λ_2; the unit circle maps to a possibly degenerate ellipse (a segment or point when an eigenvalue vanishes) with axes along the eigendirections.

Remark 9.22 (Products of observables). The product of two self-adjoint operators is self-adjoint if and only if they commute, since $(ST)^\dagger = T^\dagger S^\dagger = TS$. A product of observables is again an observable only for compatible (commuting) observables.

Example 9.23 (Expectation values are real). For a self-adjoint A and a normalized state $|\psi\rangle$ in the finite model, the expectation value $\langle A\rangle = \langle\psi| A |\psi\rangle = \langle\psi, A\psi\rangle$ is real by Theorem 9.14. The measured average of an observable is therefore a real number, exactly as physics requires. ◁

Example 9.24 (Energies are real, eigenstates orthogonal). For a finite-dimensional, time-independent Hamiltonian $\hat{H}$, the energy eigenvalues E_n are real and eigenspaces for distinct energies are orthogonal. Choose a normalized orthonormal basis inside every degenerate eigenspace; only then does $\langle E_m \mid E_n\rangle = \delta_{mn}$ hold for the chosen vectors. Equivalently, write a state as $|\psi\rangle = \sum_E P_E|\psi\rangle$ and evolve each whole energy eigenspace by $e^{-iEt/\hbar}$. Discrete pure-point models work the same way; continuous-spectrum Hamiltonians such as the free Hamiltonian require Lecture 14. ◁

Example 9.25 (A real quadratic form). For $\sigma_z = \operatorname{diag}(1,-1)$ and $v = (a,b)$, $\langle v, \sigma_z v\rangle = \overline{a}\,a - \overline{b}\,b = |a|^2 - |b|^2 \in \mathbb{R}$, illustrating Theorem 9.14: the quadratic form of an observable is always real, whatever the state. ◁

3 Unitary operators

Definition 9.26 (Unitary). $U \in \mathcal{L}(V)$ is *unitary* if $U^\dagger U = UU^\dagger = I$, i.e. $U^\dagger = U^{-1}$.

Theorem 9.27 (Unitary means inner-product preserving). For $U \in \mathcal{L}(V)$ the following are equivalent:

1. U is unitary;
2. $\langle Uu, Uv\rangle = \langle u, v\rangle$ for all u, v (preserves the inner product);
3. $\|Uv\| = \|v\|$ for all v (preserves norms);
4. U maps some (every) orthonormal basis to an orthonormal basis.

Proof. (1)⇒(2): $\langle Uu, Uv\rangle = \langle u, U^\dagger U v\rangle = \langle u, v\rangle$. (2)⇒(3): take $u = v$. (3)⇒(2): recover the inner product from the norm by polarization (Lecture 7). (2)⇒(4): if $\langle e_j, e_k\rangle = \delta_{jk}$ then $\langle Ue_j, Ue_k\rangle = \delta_{jk}$. (4)⇒(1): if U sends the ONB e_k to an ONB, then $U^\dagger U$ has matrix δ_{jk}, so $U^\dagger U = I$, and in finite dimensions a left inverse is a two-sided inverse. ∎

Theorem 9.28 (Unit-circle spectrum). Every eigenvalue of a unitary operator has absolute value 1.

Proof. If $Uv = \lambda v$ with $v \neq 0$, then $\|v\| = \|Uv\| = |\lambda|\,\|v\|$, so $|\lambda| = 1$; the eigenvalues lie on the unit circle, $\lambda = e^{i\theta}$. ∎

On a complex finite-dimensional space, time evolution $e^{-i\hat{H}t/\hbar}$ is unitary, which is why it preserves total probability $\|\psi\| = 1$; unitary symmetry operations preserve the same inner products.

Example 9.29 (Unitary operators). The rotation $R = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ satisfies $R^\dagger R = I$ and preserves lengths in $\mathbb{R}^2$. Each Pauli matrix is also unitary (besides being Hermitian), since $\sigma^\dagger\sigma = \sigma^2 = I$; indeed a *Hermitian* operator is unitary precisely when $A^2 = I$. (Self-inverse alone is not enough: $\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$ squares to I without being Hermitian.) ◁

Example 9.30 (Unit-circle eigenvalues). Over $\mathbb{C}$ the rotation R above has eigenvalues $e^{\pm i\theta}$ with eigenvectors $(1, \mp i)$: both lie on the unit circle, as Theorem 9.28 requires, and the eigenvectors are orthogonal because R is normal. ◁

Example 9.31 (Verifying unitarity). The Hadamard operator $U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ satisfies $U^\dagger U = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = I$, so it is unitary; it maps the basis $|0\rangle, |1\rangle$ to the orthonormal pair $|\pm\rangle$ of Lecture 7. ◁

Example 9.32 (Unitarity preserves overlaps). Under the same Hadamard U, $|0\rangle \mapsto |+\rangle$ and $|1\rangle \mapsto |-\rangle$, and $\langle U0, U1\rangle = \langle + \,|\, -\rangle = 0 = \langle 0 \,|\, 1\rangle$: the overlap of the images equals the overlap of the originals, as Theorem 9.27 guarantees. ◁

Remark 9.33 (Change of orthonormal basis is unitary). The matrix relating two orthonormal bases has orthonormal columns, hence is unitary (Theorem 9.27). Passing between orthonormal bases is always a unitary transformation, and an operator's matrix transforms as $A' = U^\dagger AU$ with U unitary—the inner-product-respecting refinement of the similarity $A' = P^{-1}AP$ from Lecture 4.

Remark 9.34 (Extension: generating unitaries). On a complex inner-product space, if A is self-adjoint then e^{iA} is unitary, because $(e^{iA})^\dagger = e^{-iA^\dagger} = e^{-iA} = (e^{iA})^{-1}$. This is the finite-dimensional shadow of Stone's theorem: every *strongly continuous* one-parameter unitary group has a unique self-adjoint generator. Ordinary continuity suffices in finite dimensions; unbounded generators and their domains belong to Lecture 14.

4 Normal operators

Definition 9.35 (Normal). $T \in \mathcal{L}(V)$ is *normal* if it commutes with its adjoint, $T^\dagger T = TT^\dagger$.

Self-adjoint operators ($T^\dagger = T$) and unitary operators ($T^\dagger = T^{-1}$) are obviously normal; normality is the common umbrella. Its usefulness rests on one identity.

Theorem 9.36 (Normal length identity). T is normal if and only if $\|Tv\| = \|T^\dagger v\|$ for all v.

Proof. For any T, $\|Tv\|^2 = \langle Tv, Tv\rangle = \langle v, T^\dagger Tv\rangle$ and $\|T^\dagger v\|^2 = \langle v, TT^\dagger v\rangle$. These agree for all v iff $\langle v, (T^\dagger T - TT^\dagger)v\rangle = 0$ for all v; as $T^\dagger T - TT^\dagger$ is self-adjoint, Lemma 9.13 says that holds iff it is 0. ∎

Corollary 9.37 (Shared eigenvectors with the adjoint). If T is normal, then $Tv = \lambda v$ if and only if $T^\dagger v = \overline{\lambda} v$. Consequently eigenvectors of T for distinct eigenvalues are orthogonal.

Proof. $T - \lambda I$ is normal (it commutes with its adjoint $T^\dagger - \overline{\lambda} I$), so by Theorem 9.36 $\|(T - \lambda I)v\| = \|(T^\dagger - \overline{\lambda} I)v\|$; the left side is 0 iff the right side is. For orthogonality, with $Tu = \alpha u$ and $Tv = \beta v$ we get $T^\dagger v = \overline{\beta} v$, so $\overline{\alpha}\langle u, v\rangle = \langle Tu, v\rangle = \langle u, T^\dagger v\rangle = \overline{\beta}\langle u, v\rangle$, forcing $\langle u, v\rangle = 0$ when $\alpha \neq \beta$. ∎

Example 9.38 (Normal but neither Hermitian nor unitary). $A = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ has $A^\dagger = \left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$ and $AA^\dagger = A^\dagger A = 2I$, so A is normal; yet $A^\dagger \neq A$ and $AA^\dagger = 2I \neq I$, so it is neither self-adjoint nor unitary. Its eigenvalues $1 \pm i$ are neither real nor on the unit circle—the general normal case. ◁

Example 9.39 (Checking normality by lengths). For the same A, $\|Av\|^2 = \langle v, A^\dagger Av\rangle = 2\|v\|^2 = \langle v, AA^\dagger v\rangle = \|A^\dagger v\|^2$, so $\|Av\| = \|A^\dagger v\|$ for every v—the length test of Theorem 9.36, passed. ◁

Remark 9.40 (Positive operators). For any T, the operator $T^\dagger T$ is self-adjoint and *positive*: $\langle v, T^\dagger Tv\rangle = \|Tv\|^2 \geq 0$, so its eigenvalues are nonnegative. Positive operators are the analogue of the nonnegative reals and underlie the singular value decomposition and quantum density matrices.

Example 9.41 (Orthogonal eigenvectors of a normal operator). The normal operator $A = \left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ has eigenvalues $1 \pm i$ with eigenvectors $(1, \pm i)$, and $\langle (1, i), (1, -i)\rangle = 1 + (-i)(-i) = 0$. Distinct eigenvalues give orthogonal eigenvectors, as Corollary 9.37 promises—even though A is neither self-adjoint nor unitary. ◁

Remark 9.42 (Extension: polar and singular value decompositions). Just as every complex number is $re^{i\theta}$, every operator factors as $T = UP$ with U unitary and $P = \sqrt{T^\dagger T}$ positive (the *polar decomposition*); diagonalizing P in orthonormal bases yields the *singular value decomposition* $T = U\Sigma V^\dagger$, the workhorse of data analysis and low-rank approximation. Both rest on the adjoint built here.

> **Theorem 9.43 (Cartesian decomposition).** Every operator on a complex space is $T = A + iB$ with $A = \frac{1}{2}(T + T^\dagger)$ and $B = \frac{1}{2i}(T - T^\dagger)$ self-adjoint, and T is normal if and only if A and B commute.

Proof. Both A and B are self-adjoint by direct check, and $A + iB = T$. A short computation gives $AB - BA = \frac{1}{2i}(T^\dagger T - TT^\dagger)$, which vanishes exactly when T is normal. The pair (A, B) are the operator "real and imaginary parts" of T, in analogy with $z = a + ib$. ∎

Example 9.44 (Cartesian parts of an operator). For $T = \left(\begin{smallmatrix} 1 & 2i \\ 0 & 3 \end{smallmatrix}\right)$, $T^\dagger = \left(\begin{smallmatrix} 1 & 0 \\ -2i & 3 \end{smallmatrix}\right)$, so the self-adjoint parts are

$$A = \frac{T + T^\dagger}{2} = \begin{pmatrix} 1 & i \\ -i & 3 \end{pmatrix}, \qquad B = \frac{T - T^\dagger}{2i} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

both Hermitian, with $T = A + iB$. Here $AB \neq BA$, so T is not normal—as expected for a non-diagonal upper-triangular matrix. ◁

Remark 9.45 (Extension: skew-adjoint operators). On a complex inner-product space, an operator with $B^\dagger = -B$ is *skew-adjoint*; then iB is self-adjoint, so skew-adjoint operators have purely imaginary eigenvalues. Their exponentials are unitary, which is why the generator of time evolution, $-i\hat{H}/\hbar$, is skew-adjoint—the bridge between observables and symmetries.

Example 9.46 (A skew-adjoint operator). $B = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, regarded on $\mathbb{C}^2$ (or obtained by complexifying its action on $\mathbb{R}^2$), satisfies $B^\dagger = -B$, so it is skew-adjoint with purely imaginary eigenvalues $\pm i$; indeed $iB = \left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right) = \sigma_y$ is Hermitian. Its exponential $e^{\theta B}$ is precisely the rotation of Example 9.29: a unitary generated by a skew-adjoint operator. ◁

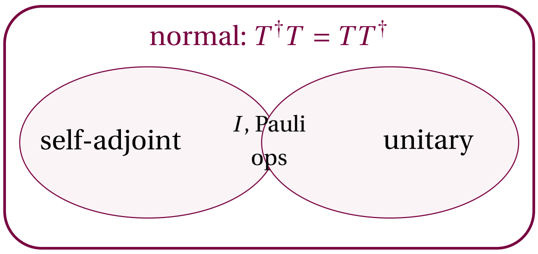

Figure 2: The self-adjoint and unitary operators are distinct families inside the normal operators; their overlap (operators that are both, such as I and the Pauli matrices) consists of self-inverse Hermitian operators.

5 Observables, symmetries, and what comes next

For the finite/discrete pure-point model, the dictionary is now complete. A self-adjoint operator plays the role of a real number: its eigenvalues are real (ideal-projective measurement outcomes) and its eigenvectors for distinct outcomes are orthogonal (distinguishable states). A unitary operator plays the role of a complex number of modulus one: it preserves all inner products and its eigenvalues are phases $e^{i\theta}$. Both are normal, and normality is exactly the property—by the finite complex spectral theorem of Lecture 10—that guarantees an orthonormal basis of eigenvectors. Multiplication by x on $L^2[0,1]$ is the standard warning:

it is self-adjoint and normal but has no nonzero ordinary eigenvectors; Lecture 14 replaces finite sums by spectral measures.

In the finite model, self-adjoint observables have orthonormal eigenbases with real eigenvalues, and closed-system evolution is unitary. Unitary symmetries preserve inner products; antiunitary symmetries are the conjugate-linear alternative. The next lecture proves that every normal operator on a finite-dimensional complex space is diagonal in an orthonormal basis.

Proposition 9.47 (Extension: uncertainty relation). For self-adjoint A, B and a unit state $|\psi\rangle$, writing $\Delta X = X - \langle X\rangle I$ for $X = A, B$,

$$\langle(\Delta A)^2\rangle\,\langle(\Delta B)^2\rangle \geq \tfrac{1}{4}|\langle[A,B]\rangle|^2.$$

Proof. $\Delta A, \Delta B$ are self-adjoint, so $\langle(\Delta A)^2\rangle = \|\Delta A|\psi\rangle\|^2$. Cauchy–Schwarz gives

$$\|\Delta A\psi\|^2\,\|\Delta B\psi\|^2 \geq |\langle \Delta A\psi, \Delta B\psi\rangle|^2.$$

The imaginary part of $\langle \Delta A\psi, \Delta B\psi\rangle$ equals

$$\tfrac{1}{2i}\big(\langle \Delta A\psi, \Delta B\psi\rangle - \langle \Delta B\psi, \Delta A\psi\rangle\big) = \tfrac{1}{2i}\langle\psi|\,[A,B]\,|\psi\rangle,$$

using $[\Delta A, \Delta B] = [A, B]$. Since $|z|^2 \geq (\operatorname{Im} z)^2$, the bound follows. ∎

Example 9.48 (Incompatible spin components). The Pauli matrices satisfy $[\sigma_x, \sigma_y] = 2i\sigma_z$, so the bound reads $\langle(\Delta\sigma_x)^2\rangle\langle(\Delta\sigma_y)^2\rangle \geq |\langle\sigma_z\rangle|^2$: two spin components cannot both be sharp unless the third has zero expectation. This is Heisenberg uncertainty in its barest form. ◁

Remark 9.49 (Compatible observables). Commuting finite-dimensional self-adjoint operators can be simultaneously diagonalized in a single orthonormal basis (Lecture 10). In this ideal model, *compatible* observables share a complete set of eigenstates and can hold definite values together; incompatible ones such as σ_x and σ_z cannot, which is the origin of the uncertainty relation.

Summary of main concepts

The adjoint $T^\dagger$, defined by $\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$, is the conjugate transpose in an orthonormal basis and obeys $(ST)^\dagger = T^\dagger S^\dagger$, $(T^\dagger)^\dagger = T$, with null $T^\dagger = (\operatorname{range} T)^\perp$. Self-adjoint operators ($T^\dagger = T$) have real eigenvalues and orthogonal eigenvectors—the observables. Unitary operators ($U^\dagger = U^{-1}$) preserve inner products and norms, map orthonormal bases to orthonormal bases, and have eigenvalues on the unit circle—the unitary symmetries and closed-system evolutions. Antiunitary symmetries are a separate conjugate-linear possibility. Normal operators ($T^\dagger T = TT^\dagger$) satisfy $\|Tv\| = \|T^\dagger v\|$, share eigenvectors with $T^\dagger$ (conjugate eigenvalues), and have orthogonal eigenvectors for distinct eigenvalues; in finite-dimensional complex spaces they are exactly the operators the spectral theorem diagonalizes in an orthonormal basis.

- **Adjoint**—$\langle T^\dagger u, v\rangle = \langle u, Tv\rangle$; the conjugate transpose in an orthonormal basis; null $T^\dagger = (\operatorname{range} T)^\perp$.
- **Self-adjoint ($T^\dagger = T$)**—finite/discrete observable model; real eigenvalues and orthogonal eigenspaces.
- **Unitary ($U^\dagger = U^{-1}$)**—closed-system evolution and unitary symmetries; antiunitary symmetries are conjugate-linear; unitary spectrum lies on the unit circle.
- **Normal** ($T^\dagger T = TT^\dagger$)—commutes with its adjoint, so $\|Tv\| = \|T^\dagger v\|$; orthogonal eigenvectors for distinct eigenvalues, the finite complex spectral-theorem class.

Exercises for this lecture are in the companion sheet `exercises-9.pdf`.